JAGMOHAN DAS

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SUMMARY

Software Engineer with 2 years of experience building production backend and full-stack applications using Python, Django, Django REST Framework, PostgreSQL, and React. Experienced in REST API development, database design, real-time systems with WebSockets, background processing with Celery, and third-party integrations. Hands-on with AI/ML and computer vision using YOLOv8, OpenCV, PyTorch, TensorFlow, and Hugging Face. Strong in backend architecture, performance optimization, debugging, and delivering end-to-end production applications.

EXPERIENCE

AvSys International India Pvt Ltd **August 2024 - Present**

Software Engineer Hyderabad, Telangana

Enterprise Full-Stack Development

- Developed and maintained full-stack enterprise applications using Python, Django, Django REST Framework, PostgreSQL, REST APIs, Bootstrap, JavaScript, HTML, and CSS.
- Designed database schemas, implemented business logic, built RESTful APIs, and delivered frontend features across end-to-end application workflows.

Defense Aviation Simulation Platform

- Engineered backend architecture for a confidential defense aviation simulation platform, designing 25+ Django models with geospatial workflows and analytical processing using GeoDjango, PostGIS, CesiumJS, and Three.js.
- Optimized spatial queries and computational logic, reducing mission-processing time from approximately 3-4 hours to 25-30 minutes, a nearly 90% improvement.

Wildlife Monitoring / Computer Vision

- Independently developed the machine learning component of a wildlife monitoring system, covering data collection, data annotation, and YOLOv8 model training.
- Trained computer vision models with Roboflow, Google Colab, and YOLOv8 for live CCTV wildlife detection, improving robustness across varied lighting, viewpoints, and night-vision scenarios.

Military Training Platform

- Built the Flutter Android application for a confidential military training platform, synchronizing Android and web workflows.
- Implemented Riverpod state management, Watch Later, viewing history, video playback, and a custom mini-player integrated with Django REST APIs.

ProMS - Internal Project Management System

- Designed, as a core developer, the Project -> Task -> QA Verification -> Bug lifecycle with role-based access control for an internal project management product.
- Implemented real-time collaboration using Django Channels and WebSockets, along with defect tracking, QA verification, and retesting workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL
Backend: Django, Django REST Framework (DRF), REST APIs, Django Channels, Celery, JWT, Node.js
Frontend: React, Flutter, Bootstrap, jQuery, HTML, CSS, Riverpod
Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite
Geospatial: GeoDjango, PostGIS, CesiumJS, Three.js
AI/ML: YOLOv8, OpenCV, PyTorch, TensorFlow, Hugging Face, Roboflow, Google Colab
Cloud and Tools: AWS, Linux, Git, Nginx, Claude, Cursor AI

PROJECT EXPERIENCE

Blood450 - Intelligent Blood Donation and Emergency Donor Matching Platform

Django, Django REST Framework, Flutter, PostgreSQL, JWT, WebSockets | Live Demo

Designed and developed an end-to-end full-stack blood donation platform independently, architecting 25+ database models, 32 REST APIs, and a 16-screen Flutter application with Django REST Framework and PostgreSQL. Built an intelligent fallback donor matching algorithm that progressively expands the search radius (5 km -> 7 km -> 9 km -> 11 km) to maximize donor availability during emergency blood requests. Implemented secure JWT authentication, Google Sign-In (Web), donor registration, profile management, emergency request workflows, notifications, WebSocket-based real-time communication, and mobile API integration. Architected the complete database schema, backend API structure, Flutter application, and deployment pipeline. The platform is designed for future LLM, RAG, and AI-assisted emergency response capabilities.

ProMS - Enterprise Project Management System

Django, Django REST Framework, PostgreSQL, Django Channels, Bootstrap | GitHub

Designed and developed a full-stack project management platform supporting Project -> Task -> QA Verification -> Bug workflows with role-based access for Project Managers, Developers, and QA teams. Implemented real-time collaboration using Django Channels and WebSockets, along with defect tracking, QA assignment, verification ownership, retesting workflows, and an enterprise SDLC-focused delivery lifecycle.

EDUCATION

Biju Patnaik University of Technology, Odisha 2020 - 2024

B.Tech, Computer Science and Engineering | CGPA: 8.6